

AWS Cloud-Based IT Infrastructure & Support Lab

Complete Project Documentation

Purpose

A practical AWS and Linux IT-support lab demonstrating cloud networking, compute, storage, security, identity, monitoring and troubleshooting.

Documentation includes

- Complete Step 1–10 procedures
- Brief explanation of every AWS service used
- Architecture diagram
- Linux commands and troubleshooting procedures
- Monitoring and CPU-alarm test
- Security configuration and evidence checklist

AWS Services & Components — Quick Reference

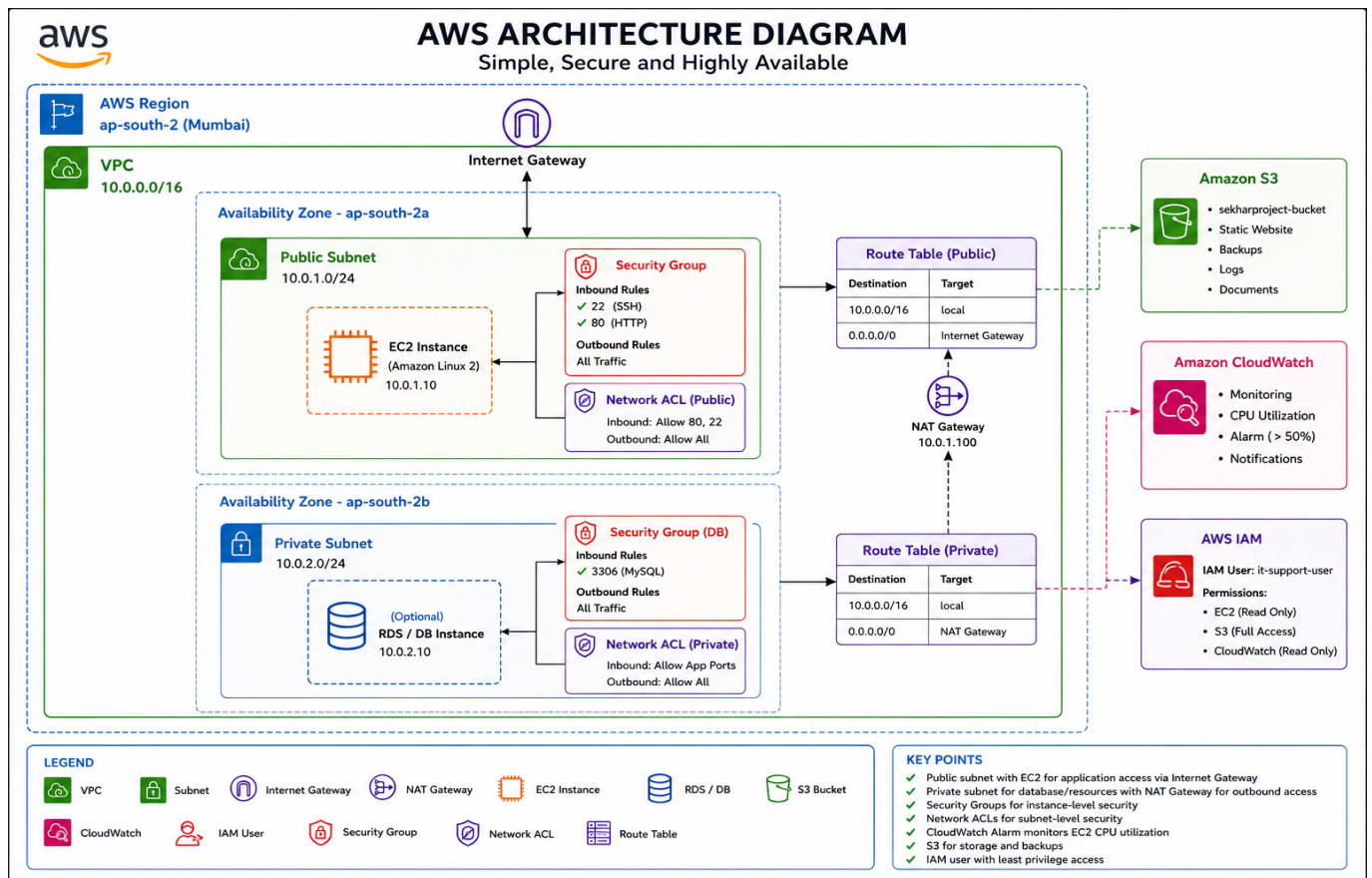
These are the main services/components used or discussed in the project. The descriptions are kept simple for project presentation and interviews.

Service	Brief explanation
AWS Region	The geographic area where AWS resources are deployed.
VPC	Virtual Private Cloud — an isolated virtual network for the project.
Availability Zone	An isolated infrastructure location inside an AWS Region.
Public Subnet	A subnet with a route toward an Internet Gateway; used for resources that need Internet connectivity.
Private Subnet	A subnet designed for resources without direct inbound Internet access.
Internet Gateway (IGW)	Connects a VPC to the Internet when routing and security rules allow it.
NAT Gateway	Allows private-subnet resources to make outbound Internet connections without accepting unsolicited inbound connections.
Route Table	Contains rules that decide where network traffic is sent.
Security Group	A stateful virtual firewall attached to resources such as EC2.
Network ACL	A stateless, subnet-level firewall that filters inbound and outbound traffic.
EC2	Amazon Elastic Compute Cloud — virtual server used for the Linux IT-support lab.
EBS	Elastic Block Store — persistent block storage used by EC2.
S3	Amazon Simple Storage Service — object storage for files/objects.
IAM	Identity and Access Management — controls users, groups, roles and permissions.
CloudWatch	Monitoring service used to observe EC2 CPU utilization and alarms.
RDS	Managed relational database service; covered as part of the database concepts in the project.
DynamoDB	Managed NoSQL database service; covered as part of the database concepts in the project.

Important: RDS and DynamoDB are included here as service concepts from the project learning/workbook. The final evidence should only claim resources that were actually created.

AWS Architecture Diagram

The diagram provides a simple visual view of the project instead of showing every AWS option. It focuses on the main network, compute, security and monitoring relationships.



How to read it: Internet traffic reaches the VPC through the Internet Gateway. Route tables decide the path. Security Groups and Network ACLs control traffic. EC2 provides compute, EBS provides its disk, CloudWatch monitors it, S3 provides object storage, and IAM controls permissions.

Project Procedures — Steps 1 to 3

Step 1 — Planning & Region

- Define the IT Support/Cloud project objective.
- Select the AWS Region for the lab.
- Plan VPC, subnets, EC2, storage, security and monitoring resources.
- Use clear resource names for easier documentation.

Step 2 — VPC, Subnets, Gateways & Routing

- Open VPC console.
- Create the custom VPC and CIDR range.
- Create Public and Private Subnets.
- Create/attach the Internet Gateway.
- Create Public and Private Route Tables.
- Associate the correct subnets with the route tables.
- Create/configure NAT Gateway when private-subnet outbound Internet access is required.
- Verify the Resource Map.

Step 3 — EC2 & EBS

- Open EC2 → Instances → Launch Instance.
- Select the Linux AMI and suitable instance type.
- Select the project VPC and public subnet.
- Attach the project Security Group.
- Configure the root EBS volume.
- Create/select the key pair.
- Launch and wait for Running status.
- Connect through EC2 Instance Connect/terminal.

Commands used

```
hostname  
ip addr  
ip route
```

Project Procedures — Steps 4 to 6

Step 4 — EC2 Verification & Linux Basics

- Connect to the EC2 terminal.
- Verify hostname, IP address and routing.
- Create/test files and check permissions.
- Check basic system information and resource usage.

Commands used

```
hostname  
ip addr  
ip route  
ls -l  
pwd
```

Step 5 — S3 & Storage

- Create the S3 bucket with a unique name.
- Keep Block Public Access enabled for the project bucket.
- Use Bucket Owner Enforced/Object Ownership where configured.
- Upload a test/project object.
- Verify the bucket and object.
- Understand EBS as EC2 block storage and EFS as shared file storage.

Step 6 — Database & Infrastructure Concepts

- Review RDS as a managed relational database service.
- Review DynamoDB as a managed NoSQL database service.
- Understand backups and snapshots.
- Understand how storage and database resources fit into a secure VPC design.
- Document only resources actually created.

Project Procedures — Steps 7 to 9

Step 7 — CloudWatch Monitoring & CPU Stress Test

- Open CloudWatch → Metrics → EC2 → Per-Instance Metrics.
- Select CPUUtilization and display the graph.
- Create/review the EC2 CPU alarm.
- Install stress-ng on EC2.
- Run controlled CPU stress.
- Observe the CPU graph and alarm.
- Stop the stress test after evidence is captured.

Commands used

```
sudo dnf install -y stress-ng
stress-ng --version
stress-ng --cpu 2 --timeout 10m --metrics-brief
Ctrl + C
```

Step 8 — Security Implementation

- Review IAM users/groups and permissions.
- Apply least-privilege access.
- Review Security Group inbound/outbound rules.
- Keep required SSH (22) and HTTP (80) rules only.
- Review Network ACL rules and subnet association.
- Review S3 Block Public Access and Object Ownership.

Step 9 — IT Support Troubleshooting

- Check network configuration.
- Test connectivity.
- Simulate a file permission problem and restore access.
- Check disk capacity and directory usage.
- Check disks and partitions.
- Check failed services and logs.
- Check memory usage.

Commands used

```
ip addr
ip route
ping
curl
chmod 000 test.txt
chmod 644 test.txt
cat test.txt
df -h
sudo du -sh /* 2>/dev/null
lsblk
systemctl --failed
journalctl -p err -n 20
free -h
```

Step 10 — Documentation & Career Evidence

- Create a clean architecture diagram.
- Organize important screenshots as project evidence.
- Write the GitHub README/project documentation.
- Upload the project and evidence to GitHub.
- Create a professional LinkedIn post describing the project and skills demonstrated.
- Add the project to the resume with concise, truthful bullet points.

Final Evidence Checklist

- EC2 instance running
- EC2 terminal / troubleshooting commands
- CPU stress and CPU utilization increase
- CloudWatch CPU alarm and graph
- Security Group configuration
- Network ACL configuration
- VPC Resource Map
- IAM user permissions
- S3 bucket/object
- Architecture diagram

Resume-Ready Project Summary

AWS Cloud-Based IT Infrastructure & Support Lab — Built a small AWS infrastructure using VPC, EC2, S3, IAM, Security Groups, Network ACLs and CloudWatch; configured networking and security controls; performed Linux troubleshooting; and tested CPU monitoring with a CloudWatch alarm.

This documentation is intended as the complete project reference. When publishing screenshots or GitHub evidence, include only resources and configurations that were actually created and tested.